CreditPathAI – Detailed Exploratory Data Analysis (EDA) Handout

This document provides an extensive guide to performing Exploratory Data Analysis (EDA) on the loan default dataset used in the CreditPathAI project. EDA is not just about plotting graphs — it is about deeply understanding the dataset, identifying patterns, uncovering hidden relationships, and preparing insights that will directly influence feature engineering and model design. A strong EDA answers: • What drives loan default? • Which features are most predictive? • Are there hidden segments of risky borrowers? • Is the dataset balanced or biased? • Are there anomalies that can mislead models? This guide walks through a comprehensive, industry-level EDA pipeline.

1. Load the Clean Dataset

import pandas as pd df = pd.read_csv("clean_loans.csv") Always perform EDA on cleaned data to ensure reliability of insights.

2. Dataset Overview

df.shape df.columns df.info() Understand number of rows, columns, datatypes, and structure.

3. Target Variable Analysis (MOST IMPORTANT)

df['default_status'].value_counts(normalize=True) Insights: • Check class imbalance • % of defaulters vs non-defaulters If highly imbalanced: → Use class weights later in modeling Visualization: sns.countplot(x='default_status', data=df)

4. Numerical Feature Analysis

Key columns: age, income, loan_amount, credit_score, dti_ratio, interest_rate df.describe() For each feature: • Mean vs median (detect skew) • Min/Max (detect outliers) Plots: sns.histplot(df['income'], kde=True) sns.histplot(df['credit_score'], kde=True) Insights to extract: • Are incomes skewed? • Do low credit scores cluster? • Are loan amounts heavy-tailed?

5. Feature vs Target Analysis (CRITICAL)

Compare distributions across default classes. sns.boxplot(x='default_status', y='credit_score', data=df) sns.boxplot(x='default_status', y='income', data=df) sns.boxplot(x='default_status', y='dti_ratio', data=df) Expected insights: • Lower credit score → higher default • Higher DTI → higher default • Lower income → higher default

6. Categorical Feature Analysis

Columns: education, employment_type, marital_status, loan_purpose sns.countplot(x='loan_purpose', hue='default_status', data=df) Insights: • Which loan purposes default more? • Does employment type affect repayment?

7. Correlation Analysis

corr = df.corr() sns.heatmap(corr, annot=True) Focus on: • correlation with default_status • multicollinearity between features

8. Feature Interactions (ADVANCED)

Analyze combinations: sns.scatterplot(x='income', y='loan_amount', hue='default_status', data=df) Insights: • High loan + low income → high risk

9. Outlier Impact Analysis

Check extreme values: sns.boxplot(df['income']) Decide: • remove or cap outliers

10. Segmentation Analysis (VERY IMPORTANT)

Group borrowers: df.groupby('education')['default_status'].mean() df.groupby('loan_purpose')['default_status'].mean() Insights: • Risk segments of population

11. Key Insights to Document

Students must write: • Top 5 factors affecting default • High-risk borrower profiles • Unexpected patterns Example: • Credit score < 600 → high default • DTI > 0.5 → risky borrowers

12. Final Deliverable – EDA Report

Report must include: • Graphs • Observations • Business insights • Data-driven conclusions This report is evaluated in Milestone 2.